1) Which of the following statements about the electric field $\boldsymbol{E}$ is correct?

A) The force acting on a charged particle is always in the same direction of the electric field.

B) The electric field experienced by a test charge depends only on its position, but not on its velocity.

C) The intensity of the electric field is always: $E = kQq/r^2$ .

D) None of the previous options is correct.

2) An electron with initial velocity parallel to the y axis: $\boldsymbol{v} = (0, v_y, 0)\ m/s$ in an electric field parallel to x axis: $\boldsymbol{E} = (E_x, 0,0)\ N/C$. The mass of the electron is $m_e$ and its charge is $-e$.

What is the final velocity ($\boldsymbol{v_f}$) of the electron after an arbitrary time $t$?

A) $v_f = \left(\frac{-eE_x t}{m_e}, v_y, 0\right)$

B) $v_f = \left(0, v_y - \frac{eE_x t}{m_e}, 0\right)$

C) $v_f = \left(\frac{eE_x t}{m_e}, v_y, 0\right)$

D) $v_f = \sqrt{\left(\frac{eE_x t}{m_e}\right)^2 + v_y^2}$

3) Which of the following statements about polarization is true:

A) Helium atoms cannot be polarized.

B) Every molecule composed of exactly two atoms is a natural dipole.

C) Carbon dioxide ($CO_2$) is not a natural dipole.

D) Dipoles can exist only in water.

4) Which of the following formulas is used to represent the short-range repulsions between atoms?

A) $U_C = \frac{q_i q_j}{4\pi\varepsilon_0 r}$ (Coulomb potential)

B) $U_{LJ} = \left(\frac{A}{r^{12}} - \frac{B}{r^6}\right)$ (Lennard Jones potential)

C) $U_\alpha = k_\alpha(\alpha - \alpha_0)^2$ (Angular harmonic potential)

D) $U_\varphi = \sum C_k \cos(\varphi - \varphi_{0,k}))$ (Potential for dihedral angles)

5) Which of the following statements about electronic orbitals is correct:

A) 1s and 2s have the same spherical symmetry.

B) 1s and 2s have the same energy.

C) 2s and 2p have the same spherical symmetry.

D) None of the above answers is correct.

6) If $\psi(x)$ represent the wave function of an electron as a function of a coordinate x:

A) $|\psi(x)|^2$ is the probability density of finding the electron along the coordinate x.

B) $\psi(x)$ is the probability density of finding the electron along the coordinate x.

C) $\psi(x)$ has always spherical symmetry.

D) $\psi(x)$ is always a gaussian function.

7） Which of the following statements about Ramachandran angles (ϕ and ψ) is correct?

A) All combinations of ϕ and ψ dihedral angles are equally probable in proteins.

B) Ramachandran angles cannot be defined for proline, because it has a different backbone compared to other amino acids.

C) Glycine and arginine have identical Ramachandran plots.

D) Ramachandran plots show that certain ϕ and ψ angle combinations are forbidden due to steric clashes.

8） Which of the following statements is true?

A) In most classical force fields, electrons are represented as clouds around the nuclei.

B) Dihedral angles in proteins are well approximated by stiff harmonic oscillators that prevent large variations of the angle.

C) In most classical force fields, the charge of each atom can only be an integer number.

D) None of the above answers is correct.

9) The energy required to bring an electron from the state 1s to the state 2s in a hydrogen atom:

A) It is much higher than the thermal energy at 300K.

B) It is comparable to the thermal energy at 300K.

C) It is lower than that required to bring the electron from the state 2s to the state 3s.

D) It is approximately equal to that required to bring the electron from the state 2s to the state 3s.

10) Which of the following statements about electrostatic energy is correct?

A) The potential energy change between two points in space does not depend on the trajectory connecting the two points.

B) The electrostatic potential energy is a vector.

C) The kinetic energy of a charged particle in an electric field never changes, because the field is always perpendicular to the velocity.

D) The formula for electrostatic energy is always $U = kQq/r$.

11) Figure 1 schematically shows how water molecules arrange around two ions (Na+ and Cl-).

a) Identify correctly which of the ions is Na+ and which is Cl-.

b) Explain what the meaning of the coefficient $\varepsilon_r$ is in the formula for the electrostatic field produced by ions in a polar solvent: $E = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{q}{r^2}$.

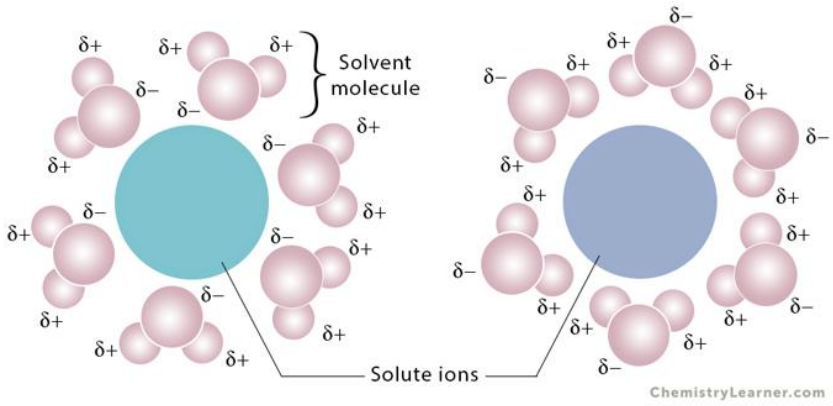



Figure 1.

12) Figure 2 shows the electronic orbitals of a water molecule. Answer the following questions and provide brief justification for your answers.

a) Which orbitals are occupied in the ground state?

b) What is the energy difference between the last occupied state and the first excited state?

c) Can electrons be found in the first excited state in a water molecule at room temperature?

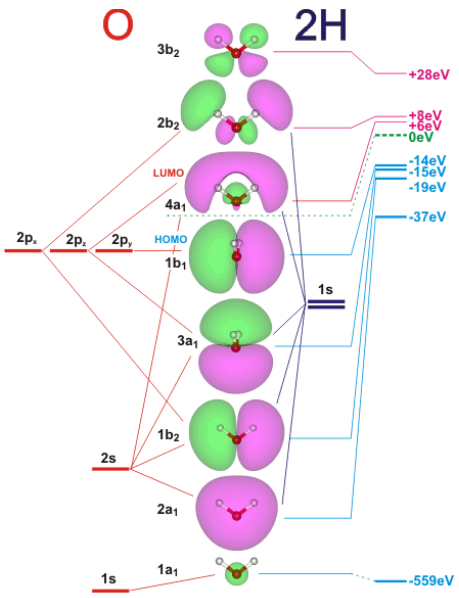



Figure 2.